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Estimating the effective reproduction number of dengue considering temperature-dependent generation intervals

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ABSTRACT

The effective reproduction number, R_t, is a measure of transmission that can be calculated from standard incidence data to timely detect the beginning of epidemics. It has being increasingly used for surveillance of directly transmitted diseases. However, current methods for R_t estimation do not apply for vector borne diseases, whose transmission cycle depends on temperature. Here we propose a method that provides dengue's R_t estimates in the presence of temperature-mediated seasonality and apply this method to simulated and real data from two cities in Brazil where dengue is endemic. The method shows good precision in the simulated data. When applied to the real data, it shows differences in the transmission profile of the two cities and identifies periods of higher transmission.

1. Introduction

As new diseases emerge more frequently and globally, traditional surveillance programs are challenged to innovate in the way they track and report risks. One of the functions of surveillance is to detect outbreaks and to initiate timely control actions (Farrington et al., 1996). Traditional protocols linking control actions to incidence thresholds are doomed to having both delays due to data collecting time and alerts issued only after time of action. The effective reproduction number, R_t, is a measurement of transmission that can be directly calculated from incidence data, whose mathematical properties within the mathematical theory of epidemics are well established (Diekmann and Heesterbeek, 2000). The dynamics of an infectious disease bifurcates at $R = 1$, with higher values implying disease spread, and lower values implying epidemic retraction or local extinction (Diekmann and Heesterbeek, 2000). Despite the long history, the routine application of R_t in a public health context as a surveillance index is recent and relies on the development of probabilistic models for estimating R_t from incidence data, when the number of cases are known but no information on who was infected by whom is available (Wallinga and Teunis, 2004; Cauchemez et al., 2006). The worth of R_t for monitoring directly transmitted infectious diseases is illustrated by Wallinga and Teunis (2004) who developed a likelihood-based method to estimate R_t from case data during the SARS epidemic and showed that despite the differences in incidence, Vietnam, Hong Kong and Canada shared similar transmission dynamics in terms of the effective reproduction number. Cowling et al. (2010) proposed a surveillance system for influenza that prospectively measured the effective reproduction number and recommend its use to guide mitigation strategies. Churcher et al. (2014) presented the usefulness of R_t for assessing the control status of malaria. The recognition of the importance of monitoring transmission has led to investments to make it more easily calculated, for example, as part of built statistical packages (Obadia et al., 2012; Cori et al., 2013).

The calculation of the reproductive number from incidence data would be straightforward if we knew exactly who-infected-whom during an outbreak, being the ratio between secondary and primary cases (Wallinga and Teunis, 2004). In the absence of such information, R_t estimation relies on a probabilistic statement based on the generation interval distribution, T_G, the time from infection of a primary case and infection of the secondary case (Kenah et al., 2008). The probability distribution for T_G provides the relative likelihood that case i has been infected by case j, given their difference in time of infection (Wallinga and Lipsitch, 2007). Since infection events are rarely observed, most inferences are based on the time interval between symptom onsets, the so called 'serial interval' (Fine, 2003; Nishiura, 2010). For directly transmitted diseases, the serial interval is estimated from case-to-case data collected in closely monitored outbreaks in households or closed populations (Vink et al., 2014). Different shapes of the generation interval distribution will impact on the calculation of the reproductive number of a disease (Wallinga and Lipsitch, 2007). We further refer the

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reader to Fine (Fine, 2003) for more detailed discussions on the several definitions of time intervals regarding infection diseases transmission.

The context of this study is the surveillance of endemic vector-borne diseases, in particular, dengue fever. Many factors contribute to make dengue dynamics complex and unpredictable at long term, among them the interaction of its four antigenically distinct viruses (DENV-1,2,3,4), capable of inducing temporary cross-immunity, and the temporal and spatial dynamics of *Aedes aegypti*, the invertebrate host. Monitoring the transmission potential of dengue is fundamental for the adequate public health response and this could be done by tracking the time–varying reproductive number of the disease (Codeco et al., 2016).

For indirectly transmitted diseases such as dengue, tracking the effective reproduction number requires a more elusive estimate of the generation (or serial) interval distribution (Boelle et al., 2008). Since transmission is mediated by a mosquito, defining who infected whom is unfeasible without resorting to expensive molecular techniques. Some authors have found indirect evidence for the serial interval of dengue measuring the typical duration of small spatial clusters. The underlying assumption is that serially transmitted infections will be geographically aggregated. Using space-time clustering techniques, Aldstadt et al. (2012) estimated in 18–19 days, the most frequent interval between successive dengue cases in Puerto Rico, while Buddhari et al. (2014) found temporal clusters of 15–17 days of duration in rural Thailand and interpreted them as the mean generation interval. More recently, Siraj et al. (2017) derived a probability distribution for the generation interval for dengue from first principles. The approach was to derive a distribution from the sum of the incubation period distributions plus the time taken by a mosquito to infect a person and vice-versa. Since the extrinsic incubation period is sensitive to temperature, being shorter at higher temperatures, the parameters of the generation interval distribution are temperature-dependent. The resulting shrink-expand of the generation interval directly affects the likelihood that case *i* has been infected by case *j*, given their difference in time of symptom onset.

In the present study, we propose a method for estimating dengue's R_t from case data taking into account the seasonal variation of the generation interval. First, we revisit the theoretical distribution for the dengue generation interval under the influence of seasonal temperature variations, constructed from first principles. We follow the same steps in Siraj et al. (2017) but with a different derivation of the timing of transmission from humans to mosquitoes and from mosquitoes to humans. Second, we modified a previous estimator for the effective reproductive number from case count data (Wallinga and Teunis, 2004) to include this time-varying generation interval. We apply the new estimator to simulated data under different seasonal scenarios to illustrate the importance of taking into account the seasonality on the estimation of R_t for temperature-sensitive diseases. Then, we estimate the effective reproductive number of dengue from dengue notification data for two Brazilian cities with distinct temperature regimes where R_t has been used to detect transmission in a surveillance program.

2. Methods

2.1. The transmission cycle of dengue fever

Fig. 1 shows the sequence of four steps that characterize the transmission cycle of dengue. The distribution of the generation interval was computed from first principles, as the sum of the distribution of the time taken to complete each step, a method that has been applied before for malaria (Huber et al., 2016) and dengue (Siraj et al., 2017). Here, we propose an alternative distribution to the one proposed by Siraj et al. (2017), using gamma and exponential distributions instead of lognormal distributions, and relying on different assumptions regarding the steps of transmission between vector and human and vice-versa.

We start at the point where a person is inoculated with DENV load from dengue-infected mosquitoes, mainly *A. aegypti* (Lambrechts et al.,

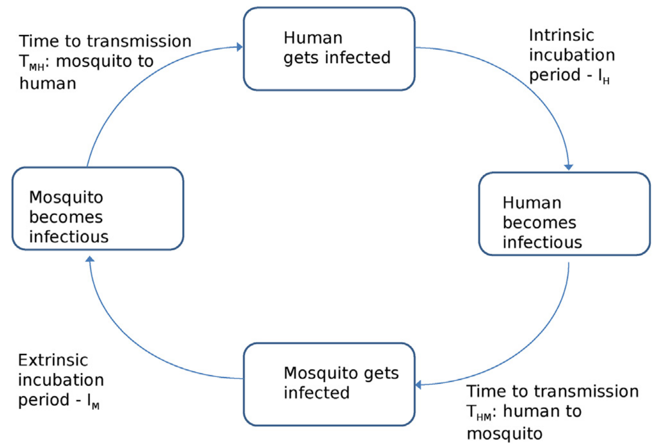

Fig. 1. Transmission cycle of dengue, divided into four steps.

2010). Starting from this point, the first step of the transmission cycle is the latent period, defined as the time from infection to infectiousness during which the virus invades the immature dendritic cells in the human skin and, through a cascade of events, reproduces and disseminates systemically through the lymphatic system. As the latent period is difficult to observe, in general, the intrinsic incubation period (I_H), the time from infection to symptoms onset, is used as an approximation (Chan and Johansson, 2012). Chan and Johansson (2012) carried out a detailed review of the literature on the intrinsic incubation period of dengue, including earlier studies where individuals were experimentally exposed to infected mosquitos and more recent studies where the time to symptom onset among travellers provided left-censured incubation period data. Fitting bayesian uncensored and censored time-to-event models, they found that either a lognormal or a gamma distribution, with expected value (mean) of 5.9 days (95% CI 3.4–9.2 days) are good models for the distribution of dengue's intrinsic incubation period (I_H) (Table 1, Fig. 2A). Here, we chose the gamma distribution for I_H for mathematical convenience, while a lognormal distribution was preferred by Siraj et al. (2017). A study by Duong et al. (2015) suggests that the latent period can be up to two days shorter than the incubation period.

A person leaves the latent phase when the viral load in the lymphatic system is sufficiently high to produce an infectious blood meal. Nguyen et al. (2013) performed experimental infections in which mosquitoes were allowed to feed on dengue-infected patients at their first, second, etc., day of illness. Probability of infection to mosquitoes was highest during the first 3–4 days from the fever onset and decayed to zero in ca. 7 days. Nguyen et al. (2013) modeled the risk of human-to-mosquito transmission per day of illness as the logistic hazard function: $h(t) = (1 + exp(-7 - \log(0.23))^{-1}$. According to this function, the risk of transmission is ca. one during the first three days of illness and decays to ca. 50% at day five, becoming negligible after day seven. From this function, we computed the probability distribution of “the time from symptom's onset to infecting a mosquito”, $f(T_{HM})$, by transforming the hazard function into a probability function using the relationship between survival and hazard functions (Sá Carvalho et al., 2011):

$$f(T_{\mathrm{HM}}) = \frac{e^{-t}}{(1 - e^{7+\log(0.23)t})^{(1-1/(-\log(0.23))}} \tag{1}$$

For $t \leq 3$ days, the denominator of equation 1 is close to 1, which corresponds to an exponential distribution with $\lambda_{TM} = 1$ which is chosen to represent the probability distribution of “the time from symptom's onset to infect a mosquito” (Fig. 2B).

The next two events in the transmission cycle describe the natural history of the dengue virus within the invertebrate host. After being ingested, these viruses initially infect the epithelial cells that line the

Table 1
Probability distributions of the four steps of the transmission cycle of dengue.

Parameters	Variable/Type of distribution	Description	Values
a_H, s_H^{-1}	Intrinsic incubation period/Gamma distribution	Shape and rate parameters, respectively	$a_H = 16$, $s_H^{-1} = 2.7$ (day^{-1})
a_M, $s_M^{-1} = b_0 - b_1 T$	Extrinsic incubation period/Gamma distribution	Shape and scale parameter as function of temperature	$a_M = 4.3$, $b_0 = 7.9$, $b_1 = -0.21$ (day^{-1})
λ_{TM}	Transmission from mosquitoes/Exponential distribution	Exponential parameter	1 (day^{-1})
λ_{TH}	Transmission from humans/Exponential distribution	Exponential parameter	1 (day^{-1})

mosquito's midgut. For a few days, they will multiply and infect neighboring cells causing a large focus of viral particles that will eventually disseminate to other tissues in the abdomen, thorax, head, neurons, and salivary glands, causing a systemic infection (Kramer and Ebel, 2003). The time between virus ingestion and presence of viruses in the saliva defines the extrinsic incubation period (I_M) which is strongly affected by air temperature. Biologically, as temperature increases (within the biological limits for the species), the replication rate of the virus within the mosquito accelerates and, by consequence, a shorter I_M is observed (Fig. 2C). Chan and Johansson (2012) present the most comprehensive analysis of available evidence on the temperature-dependent probability distribution of I_M. From a variety of studies with censored and uncensored data, they tested different probability distributions for I_M having temperature as a covariate affecting the scale parameter (Table 1). They found the best fitted probability distribution to be lognormal, followed by gamma with a small difference in performance. For mathematical reasons, we opted for the gamma distribution, without loss of precision, since both distributions present qualitatively similar shapes. The lognormal distribution is used by Siraj et al. (2017).

Once dengue viruses reach the salivary gland, the extrinsic incubation period ends and mosquitoes enter the infectious phase, remaining so throughout the rest of their lives. Mosquitos bite humans at a high rate (0.6 to 2 per day according to Scott et al. (2000)), thus it is reasonable to assume that the virus will be transmitted to humans soon after transmission conditions are attained. For how long it will continue to be transmitted is uncertain. A field study in Rio de Janeiro estimated in 7.5 to 9 days, the average longevity of *A. aegypti*, with no difference between summer and winter (Villela et al., 2015). Since most of the lifetime of an infected mosquito would be passed in the incubation phase, we assumed that mosquitos would have only a few days to infect humans. Using this argument, the 'time to infect a human' ($f(T_{MH})$) was tentatively modeled as an exponential function with parameter $\lambda_{TH} = 1$ day, resulting in most of the transmission activity occurring in the first 3 to 4 days of infectiousness (Fig. 2D). Our model contrasts with Siraj et al.'s (2017) who assumed an average of ca. 8 days for the mosquito-to-human transmission probability distribution.

In summary, the probability distributions of the four time intervals of dengue's transmission cycle were described by gamma distributions, being the exponential distribution a special case of a gamma distribution (Table 1). These random variables are assumed mutually independent.

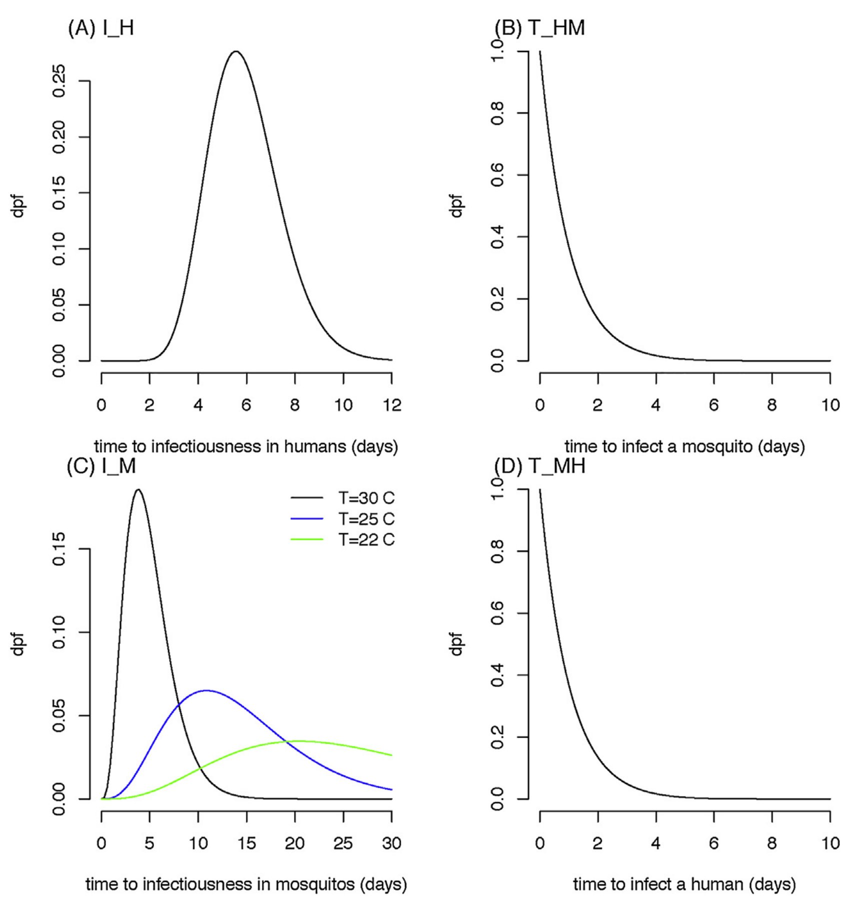

Fig. 2. Probability density functions for the (A) the intrinsic incubation period of dengue in humans, (B) the time to infect a mosquito, (C) the extrinsic incubation period in mosquitoes at 22, 25 and 30 °C, (D) time to infect a human (time unit: days).

In this formulation of the generation time, effects of temperature on mosquito's demography (Yang et al., 2009) and behavior were not considered, neither the effect of other exogeneous variables, such as humidity, that can further interact with temperature to affect mosquito competence (Smith, 1975; Simoes et al., 2013).

2.2. The generation interval distribution for dengue, under constant temperature

We start deriving an expression for the probability distribution for the dengue's generation interval, described by a random variable X_w, under a constant temperature w. By construction, X_w is the sum of the four previously defined random variables:

$$X_w = I_H + T_{HM} + I_M|w + T_{MH} \tag{2}$$

The moment generating function (MGF) for a random variable G is defined as $M_G(\theta) = E[e^{\theta G}]$. For a Gamma distribution with generic parameters a and s, the MGF is given by $M_G(\theta) = (1 - s\theta)^a$. The MGF of a sum of independent random variables is the product of MGFs of each of the variables. This property is often convenient for finding the distribution of a sum of random variables. Here, using this property, we obtain the MGF of the dengue generation time as $M_{TG}(\theta) = M_{TM}(\theta) M_{IM}(\theta, T)M_{HM}(\theta)M_{MH}(\theta)$, where M_{IM}, M_{IH}, M_{HM}, M_{MH} are moment generating functions of each of the incubation and transmission times I_M, I_H, T_{HM}, T_{MH}, given by:

$$\begin{aligned} M_{\mathrm{IM}}(\theta, T) &= (1 - s_m(w)\theta)^{-a_m},\\ M_{\mathrm{IH}}(\theta) &= (1 - s_h\theta)^{-a_h},\\ M_{\mathrm{HM}}(\theta) &= \left(1 - \frac{\theta}{\lambda_{\mathrm{HM}}}\right)^{-1},\\ M_{\mathrm{MH}}(\theta) &= \left(1 - \frac{\theta}{\lambda_{\mathrm{MH}}}\right)^{-1}. \end{aligned} \tag{3}$$

For the probability distribution $P(X_w \le t)$, let $g_w(t)$ be the probability density function. The moment generating function $M_{TG}(\theta)$ permits us to find $g_w(t)$. Since the scale parameters s_h and s_m are different, obtaining analytically an inversion of the resulting moment–generating function is not trivial, even if the temperature is considered constant. Moschopoulos (1985) described a computationally efficient numerical method for obtaining the probability density function of a variable given by a sum of multiple gamma random variables, in which coefficients are recursively defined. We implemented this numerical method to compute the dengue's generation interval distribution from the distributions defined in Table 1.

Fig. 3 shows the probability distribution for the dengue generation time, X_w, under different temperatures, assuming them to be constant. The median X_w is 4 weeks at 25 °C, but it can be as low as 2 weeks. The 10–19 days reported by Aldstadt et al. (2012) is within the 95% interval of the distribution.

2.3. Truncating the generation interval distribution

The generation interval is the serial interval between two human infections. By definition, this interval only exists if there are mosquitoes capable of acting as vectors carrying the virus from one human host to the other. Some authors have suggested to truncate the maximum lifetime of an infectious mosquito at 30 days or 40 days (Boelle et al., 2008; Huber et al., 2016). Here, we truncate the generation interval at 5 weeks, which is the sum of an estimation of the maximum longevity of mosquitoes in the field (30 days) and the average incubation period in humans (6 days) (Scott et al., 2000). This is the horizontal line in Fig. 3.

Another restriction to the generation interval distribution is imposed by low temperature. *A. aegypti* biological limit is around 15 °C which is the temperature below which oviposition and biting are halted (Vezzani et al., 2004). Models suggest that the minimum temperature required for dengue transmission is 18 °C (the confidence interval went down to 12 °C) while the maximum temperature is 32 °C (confidence interval going up to 38 °C) (Mordecai et al., 2017). According to Fig. 3, at temperatures below 18 °C, most the probability distribution would be above the biological limit of 5 weeks.

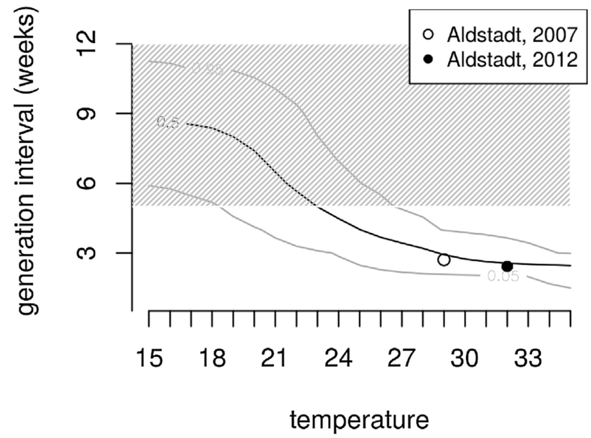

Fig. 3. Probability distribution of dengue's generation interval as a function of temperature (°C). The dashed area indicates the range of values (above 5 weeks) which are unlikely to be observed in the field due to the limitation posed on the generation interval by the mosquito maximum lifespan. The dots indicate empirical estimates of the generation interval. Since the original studies lacked information on temperature, the average summer temperature was used. Open circle: dengue epidemic in Florida, Puerto Rico, 1991 (Aldstadt, 2007); Closed circle: dengue in Kamphaeng Phet, Thailand (Aldstadt et al., 2012).

2.4. Dengue's generation interval when temperature varies

Now, we formalize an approach to derive the generation interval distribution of dengue in a scenario of varying temperature. We assume that temperature varies in discrete steps, since mostly available meteorological data are discrete measurements of temperature averaged per day or week.

Let $\mathscr{T} = \{t_0, t_1, t_2, t_3, ...\}$ determine times for temperature transitions. Let $\mathscr{S}$ be the series of temperature values $w_0, w_1, w_2, ...,$ such that $w(t) = w_i$, if $t_i \le t < t_{i+1}$. We want to obtain the distribution of generation interval considering these temperature transitions.

For a generation starting at time τ, the generation interval distribution will depend on the temperature values observed from that moment τ onwards. That is, we need to consider $P(X(\tau) \le t) = \int_0^t g(\tau, \psi)d\psi$, where now the generation interval distribution is not time-invariant, requiring a more general function $g(\tau, t)$. If under a constant temperature w the random variable X_w was required, here we have a more general distribution for $X(\tau)$ to be $P(X(\tau) \le x|w(\tau), w(\tau+1), w(\tau+2), ...)$. The probability of the generation interval to exceed a long period of time after τ is small, hence after some time the contribution of the temperature vanishes.

As the temperature changes, the generation might speed up or slow down, depending on the temperature values. In order to obtain the distribution for $X(\tau)$, the procedure is to apply piece-wise functions using the transition times as break points, assuming constant temperature between transition times. The distribution function changes on every transition time by advancing a distribution of generation interval depending on the temperature. For any time t, let us assume that we can find the probability for the most recent transition i, such that $t_i \le t < t_{i+1}$. Then, $P(X(\tau) \le t_i - \tau) = p_i$. We can map the probability to the time domain, by an inverse function, assuming a constant temperature $T_i = w_i$. Therefore, we find $t^* = F_{w_i}^{-1}(p_i)$. Since we start with t_0, having $P(X(\tau) \le t_0 - \tau) = 0$, the distribution can be obtained recursively by this process.

We proceed the analysis considering that the generation interval is ahead of time t_i given a temperature w_i, by deriving the time difference $t' = t - t_i$. For $P(X(\tau) \le t|t_i - \tau \le X < t_{i+1} - \tau)$, we sum the

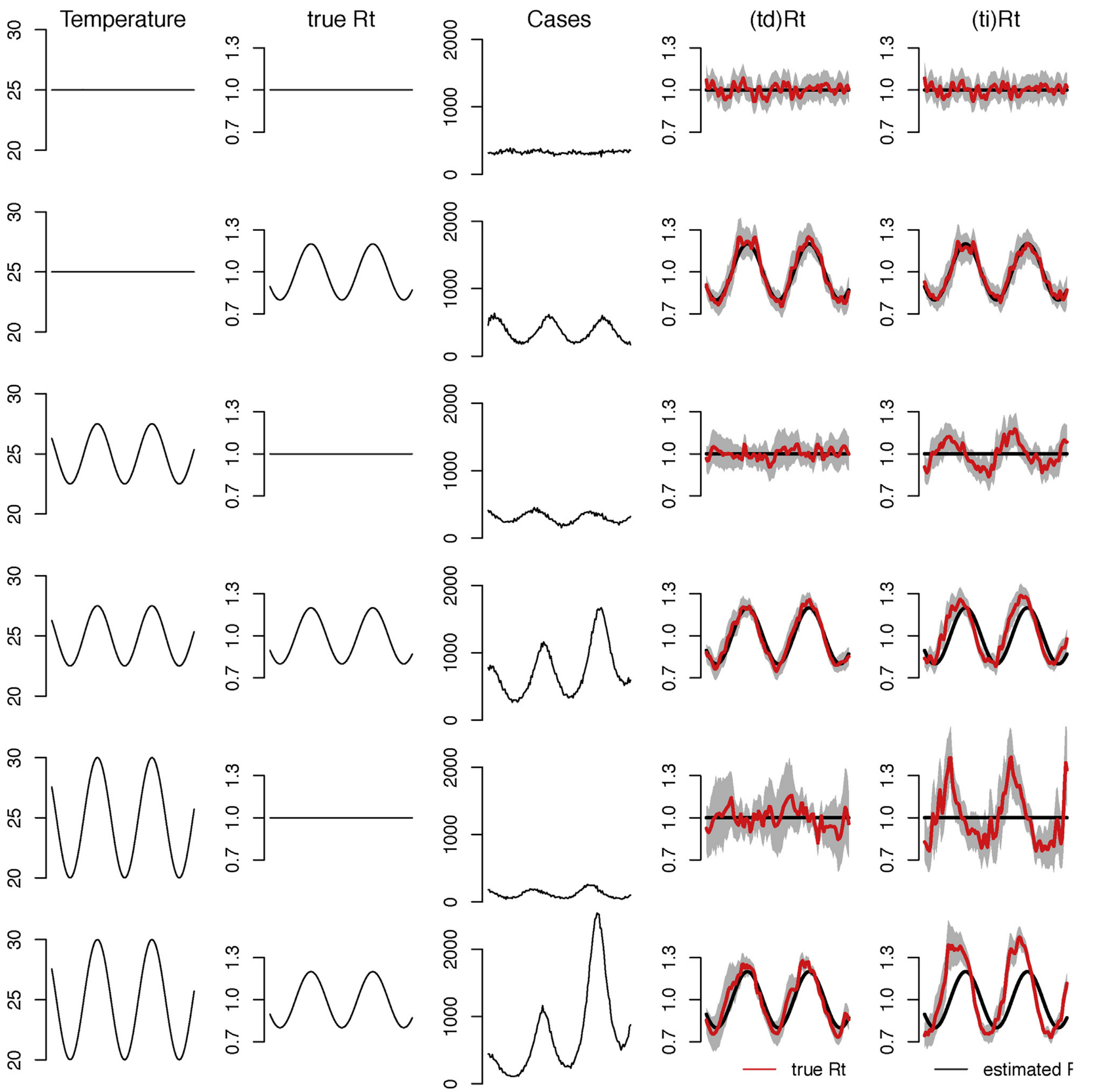

Fig. 4. Each line (top-to-bottom) corresponds to a simulated scenario drawn from a branching process model. The model describes the reproduction of a disease at steady-state (endemic phase), which has a generation interval distribution that depends on temperature (X_w (Temperature)), as described for dengue in the text. The true reproductive number used in the simulations is either constant (R_t = 1 in lines 1, 3 and 5) or slightly sinusoidal (lines 2, 4, 6). The third column shows the incidence time series generated by each scenario. The last two columns show the true and estimated R_t for each scenario obtained using the temperature-dependent estimator ((*td*) R_t, column 4) and temperature-independent estimator ((*ti*) R_t, column 5). The scenarios are: **Line 1.** Scenario with constant X_w (Temp = 25 °C), and constant transmission (R_t = 1); **Line 2.** Scenario with constant X_w (Temp = 25 °C) and seasonal transmission; **Line 3.** Scenario with seasonally varying generation interval X_w (25 ± 2.5 °C) and constant transmission; **Line 4.** Scenario with seasonally varying generation interval X_w (25 ± 2.5 °C) and seasonal transmission; **Line 5.** Scenario with seasonally varying generation interval X_w (25 ± 5 °C) and constant transmission; **Line 6**. Scenario with seasonally varying generation interval X_w (25 ± 5 °C) and seasonal transmission.

probability $P(X(\tau) \le t_i - \tau) = p_i$ and a probability given by integrating the probability density over the additional time after t_i, considering a temperature w_i. Now, we have

$$P(X(\tau) \le t | t_i - \tau \le X < t_{i+1} - \tau) = \int_0^t g(\tau, \psi)d\psi = \int_0^{t_i-\tau} g(\tau, \psi)d\psi + \int_{t_i-\tau}^t g_{w_i}(\tau, \psi)d\psi = p_i + \int_{t_i}^t g_{w_i}(\tau, \psi)d\psi, \quad (4)$$

where the last term is obtained with a constant temperature w_i.

To construct the dengue generation interval distribution from a time series of temperature from a real city, first a smooth function is applied to the series to reduce high frequency variability (optional). Then, the probability distribution of the generation interval is calculated for each data point of the temperature time series between t_0 and t_{max}, considering the generation interval up to a maximum of $GTmax$ = 35 days (or 5 weeks). Finally, $g(\tau, t)$, is calculated in the form of a matrix G given by elements $a_{i,j} = P(t_j - t_i \le X(t_i) < t_{j+1} - t_i)$, where $t_j = t_i, t_{i+1}, t_{i+2}, \ldots, t_{GTmax}$.

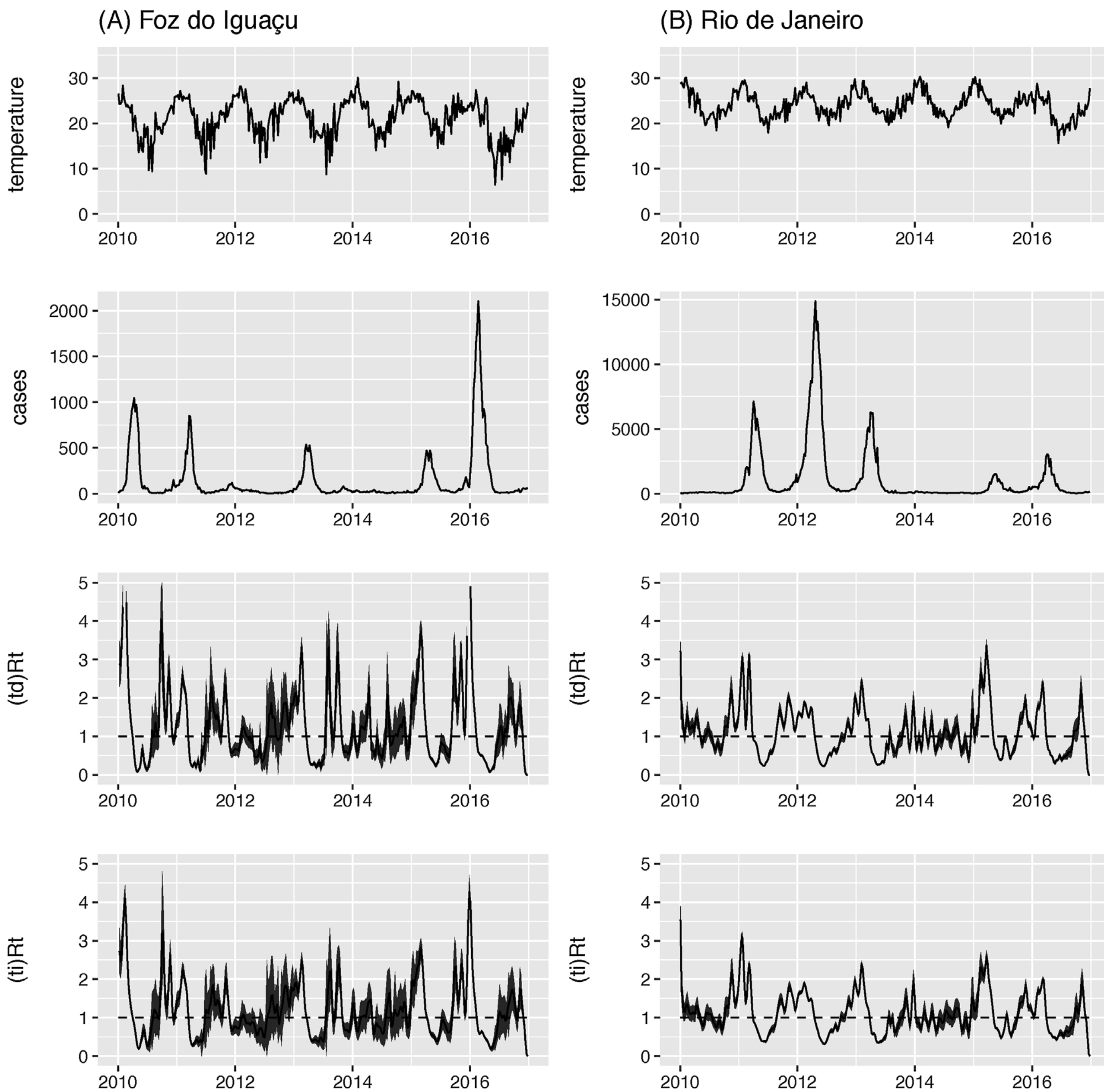

Fig. 5. Dengue and temperature data for two cities in Brazil, (A) Foz do Iguaçu and (B) Rio de Janeiro: (top) Mean weekly temperature from 2010 to 2016, (second line) number of reported cases of dengue per week; (third line) estimated R_t using the temperature-dependent estimator; (bottom line) estimated R_t using the temperature-independent estimator.

2.5. Reproduction number with temperature-dependent generation interval

Wallinga and Lipsitch (2007) formalized the estimation of R_t from case count $c(t)$ when the generation interval distribution is time-invariant. In their notation, the probability density is $g(t)$. In our framework, this notation is generalized to $g(\tau, t)$, for a generation interval starting from time τ.

$$R_t = \int_t^\infty g(t, x-t)\frac{c(x)}{\int_0^\infty c(x-a)g(x-a, a)\mathrm{d}a}\mathrm{dx}. \tag{5}$$

The Rt estimation using function $g(\tau, t)$ gives us the temperature-dependent (*td*) R_t, whereas the time-invariant function $g(t)$ gives us the temperature–independent (*ti*) R_t. The latter is already available in Package R0 (Obadia et al., 2012) in the R platform, which follows the approach described by Wallinga and Teunis (2004).

We implemented the (*td*) R_t estimator as an extension of the function available in the R0 package. The new function uses the number of new cases per unit of time and the previously defined matrix G, describing the time-varying generation time distribution. Assuming that cases at time t_i have probabilities p_j, $1 \le j \le G_{max}$, of generating secondary cases during interval (t_j, t_{j+1}) given by $\frac{a_{i,j}c_i}{\sum_{l=0}^{G_{\max}} c_{x-l}a_{x-l,l}}$, we obtain samples of secondary cases from a multinomial distribution using this set of probabilities. This procedure permits us to obtain quantiles, hence to find uncertainty intervals. This function and all data used in this study are available at https://github.com/AlertaDengue/publication-data/tree/master/Codeco-et-al-2017.

2.6. Simulated data

A simple simulation model was created to compare the performance

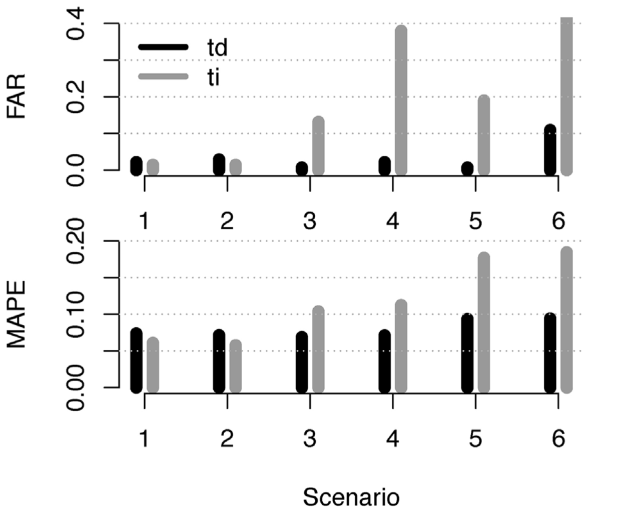

Fig. 6. Performance of the temperature-dependent (*td*) R_t and temperature-independent (*ti*) R_t estimators when used for estimating the reproductive number of dengue under the six scenarios described in Fig. 4: (top) FAR = False Alert Rate; (bottom) MAPE = mean absolute percentage error.

Table 2
Description of the two cities, Foz do Iguaçu and Rio de Janeiro: climate and dengue, from 2010 to 2016. Time unit is week.

	Foz do Iguaçu	Rio de Janeiro
Mean temperature (range)	6.5°–30°	15.6°–30°
Minimum temperature (range)	2.7°–24°	14.1°–25°
Weeks with mean temperature < 14 °C	7%	0%
Weeks with mean temperature between 14 and 18 °C	15%	0.8%
Weeks with mean temperature > 18 °C	78%	92%
Dengue counts (median, range)	34 (2–2102)	204 (17–14,870)
Dengue incidence (median, range)	12.8 (0.75 − −796) ÷ 10^5	3 (0.26 − −228) ÷ 10^5

of the temperature-independent (*ti*) and temperature-dependent (*td*) R_t estimators in scenarios where the generation interval vary seasonally or not. The simulator is a branching process which generates n_g infectors at each generation g according to the algorithm:

1. Initial conditions: n_0 infected individuals are assigned to generation 0, all of them with time-of-infection set to day 0.
2. Each infected person belonging to $g = 0$ will infect $y_0 \sim Poisson$ ($R_t(g = 0)$) persons to produce the first generation of cases, n_1, where $R_t(g = 0)$ is the reproduction number at time $t = 0$. The total number of cases in generation $g = 1$ is $n_1 = \Sigma y$.
3. A 'day-of-infection' is assigned to each of the new n_1 cases: $t_y(g = 1) = t_{p,y} + \delta$ where $t_{p,y}$ is the time-of-infection of the infector and δ is a random variable following $g(\tau, t)$.

This process is iterated for 50 generations. When finished, a weekly time series is obtained from the individual life histories by aggregating the cases in time windows of seven days which is the standard time unit used for surveillance. The model is useful for simulating scenarios close to the steady state, since the number of susceptibles does not vary. Without seasonality, at steady-state, the reproductive number is by definition equal to 1, and that is the value we want to find using the R_t estimators. We also simulated a periodic environment (Bacaër, 2007) where transmission has a small seasonal fluctuation of the form ($R_t = 1 + a\ cos(t/365)$). The model was used to generate 6 sets of incidence data (Fig. 4) varying the intensity of the temperature seasonality and the presence/absence of seasonal transmission (a).

The simulated data was used to compare the (*ti*) R_t and (*td*) R_t estimators in terms of their mean absolute percentage error (MAPE):

$$\text{MAPE} = \frac{\sum |\overline{R_t} - R_t^*|}{n} \tag{6}$$

where $\overline{R_t}$ is the estimated R_t and R_t^* is the true R_t. Since estimations were accompanied by confidence intervals, we further tested how often these estimators could be raising false epidemic alarms by wrongly detecting $Rt > 1$. To do this, we computed the proportion of time points in which the prediction interval was above 1, issuing a false alarm (FAR).

$$\text{FAR} = \frac{\sum \text{ll}(\overline{R_t}) > R_t^*}{n} \tag{7}$$

where $ll()$ is the lower limit of the 95% prediction interval, and n is the length of the time series.

2.7. Dengue incidence data

We compared the two estimators of dengue's R_t using real notification time series from 2012 to 2016 for two cities in Brazil, Rio de Janeiro and Foz do Iguaçu, where dengue is endemic. **Foz do Iguaçu** (25.5163° S, 54.5854° W) is located at the border between Brazil, Argentina and Paraguay. In 2016, the estimated population was 256 thousand inhabitants (IBGE). The Koppen climate type is Cfa (temperate without dry season and hot summer). **Rio de Janeiro** (22.9068° S, 43.1729° W) is a coast city with 6.3 million inhabitants. The Koppen climate is Am (tropical monsoon).

Notification data as collected by the Brazilian national disease notification system (SINAN) was made available to the authors through the Infodengue project (http://www.dengue.mat.br). The majority of dengue cases are confirmed using clinical-epidemiological criteria only. The criteria in SINAN are given by clinical evaluation on a set of symptoms listed by the Federal Health Ministry: fever with duration up to seven days, headache, myalgia, retro-orbital pain, arthralgia, pruritus or exanthema, possibly with hemorrhage, with the patient being in an area with dengue virus transmission in previous 15 days or with presence of *A. aegypti* mosquito. Starting in 2015, cases were separated from either Zika or chikungunya by distinguishing more significant symptoms (exanthema or arthralgia) and occurrence of transmission of ZIKV or CHIKV. Cases are confirmed once patient illnesses do not depart to other clinical manifestations.

Daily temperature for 2012–2016 at the city airports (SBGL in Rio de Janeiro and SBFI in Foz do Iguaçu) were obtained from INMET (http://www.inmet.gov.br). Mean temperature per week was used for estimating R_t (Fig. 5).

3. Results

3.1. Synthetic scenarios – simulated data

Fig. 4 shows the six time series of simulated cases generated under different levels of temperature seasonal oscillation. Lines 1, 3 and 5 show scenarios where temperature affects only the generation interval but not the transmission. These are fictitious scenarios that are useful for isolating the effect of the shrink-expand generation interval on the incidence of the disease which takes the form of an oscillation in the incidence time series in synchrony with the temperature variation. The larger the temperature variation, the larger the dengue oscillation that is produced in the data that is caused despite the invariance of the reproductive number. Lines 2, 4 and 6 show the same three temperature profiles, now in the presence of a small variation of the reproductive number. This very small variation is sufficient to induce large variations in the incidence curve, an example of resonance (Greenman and Pasour, 2011).

Fig. 4, 4th and 5th columns, shows the estimated R_t for each of these six scenarios using the proposed temperature-dependent $(td)R_t$ and the standard temperature-independent $(ti)R_t$ estimator. When temperature is constant (lines 1 and 2), both estimators behave very similarly, with (ti) R_t presenting slightly better results in terms of MAPE and FAR (Fig. 6). When the generation interval varies as a function of temperature (lines 3 to 6), the standard $(ti)R_t$ performs poorly by wrongly attributing all dengue incidence variation to variation in transmission. The result is a high percentage of false alarms ($R_t > 1$) that increases as temperature seasonality is stronger. By considering the effect of temperature in the generation interval, $(td)R_t$, on the other hand, provides a more precise and accurate estimate of the reproduction number.

3.2. Estimation of reproduction numbers in Brazilian cities

Generation interval distribution of dengue in Foz do Iguaçu and Rio de Janeiro. Table 2 and Fig. 5 describe the temperature and the incidence of dengue in the two cities during the 2010–2016 period. In Rio, temperature was always above 15 °C, meaning that *A. aegypti* was capable of reproducing the whole year around. In 92% of the weeks in Rio, conditions were further suitable for transmission (between 18 and 30 °C) (Mordecai et al., 2017). In Foz, the climate is colder, and in 7% of the weeks, temperature reached values below 15 °C. In about one quarter of the weeks, temperature was not suitable for dengue transmission. The two cities are very different in size, although the number of cases was much larger in Rio, in terms of incidence, Foz do Iguaçu reached the highest values.

Fig. 7 shows the distribution of the generation interval of dengue in Foz do Iguaçu and Rio de Janeiro, from 2000 to 2016, using mean week temperature as input for the X_w function. It is a graphical representation of the G matrix. The predicted median generation interval for dengue in Foz do Iguaçu varied from 3.0–4 weeks in the summer to more than 10 weeks in the winter. Considering that only intervals up to 5 weeks are biologically feasible, dengue transmission in Foz was possible in only 51% of the weeks, according to this metric. In Rio, the median generation interval varied from 2.5–4 weeks in the summer and 6–7 weeks in the winter. Considering the truncation at 5 weeks, conditions for transmission in Rio existed in 84% of the weeks.

Effective reproduction number. Fig. 5 presents the estimation of the $R(t)$ for the two cities, from 2012 to 2016, using the two estimators, $(td)R_t$ and $(ti)R_t$. Using $(td)R_t$, in Foz do Iguaçu, median R_t was 1.05, reaching $R_t > 4$ in three epidemic years. In weeks with temperature below 18 °C, the average R_t was 0.91, increasing to $R_t = 1.2$ at temperature between 18 °C and 24 °C, and $R_t = 1.7$ at periods with mean temperature above 24 °C. In Foz, R_t was found significantly above 1 in 33% of the weeks between 2010 and 2016, distributed through the whole year except May and June when $R_t < 1$ always (Fig. 8).

In Rio de Janeiro, median R_t was 1.02, reaching maximum of 2–3 in epidemic years. Dengue transmission is more homogeneously distributed along the year in Rio than in Foz. Between 18 and 24 °C, the median R_t was 0.82 while at temperature above this threshold, median $R_t = 1.44$. In Rio, R_t was found to be significantly above 1 in 40% of the weeks.

The temperature independent estimator tended to overestimate R_t in both cities when values where low (up to ca. 1.5) and underestimate R_t when values were higher. In average, the difference between the two estimators was not significant (0.01 in Foz and 0.02 in Rio), but for 14 times in Foz, the difference was greater than 1 and in 6 times in Rio, the difference between the two estimates was greater than 0.5.

4. Discussion

This study proposes a method to estimate dengue's real-time reproductive number from case data taking into account the seasonality of the generation interval. The function is written as an extension for the TD function in R0 package of the R environment (Obadia et al., 2012) and can be directly applied to case counts produced by national surveillance programs.

The proposed distribution for the generation interval of dengue is consistent with the available, but scarce, empirical data. Aldstadt, Aldstadt (2007), Aldstadt et al. (2012) proposed to use the serial interval between dengue cases in spatially clustered cases as a proxy for the generation interval. In Kamphaeng Phet, Thailand, where temperature varies between 25 and 35 °C, he found a strong temporal cluster of 15–17 days which he interpreted as the most frequent serial interval. He also detected weaker but still significant clusters of 19–27 days, and 30–34 days. The generation interval distribution derived here is consistent with his findings. The median generation time, according

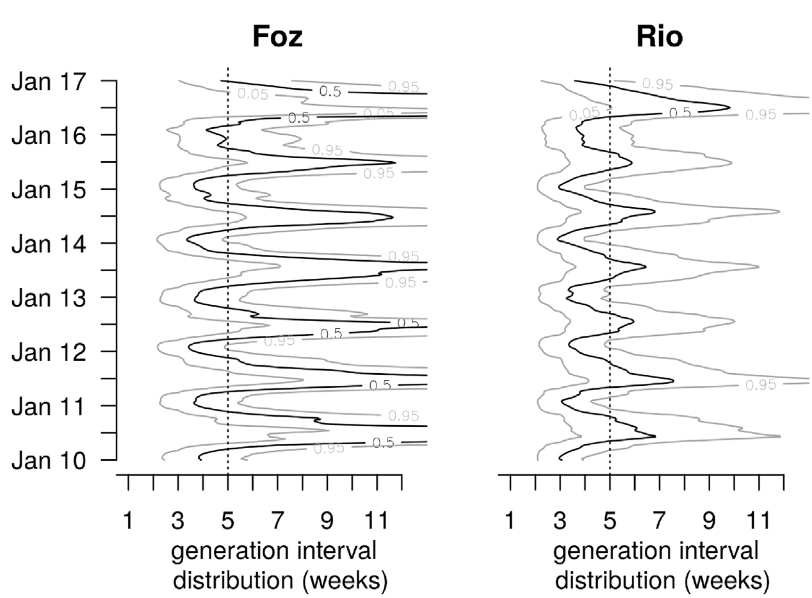

Fig. 7. Distribution of dengue's generation interval in Foz do Iguaçu and Rio de Janeiro, from 2010 to 2017. The curves show the percentiles.05,.50 and.95. The vertical line marks the maximum biologically feasible generation interval.

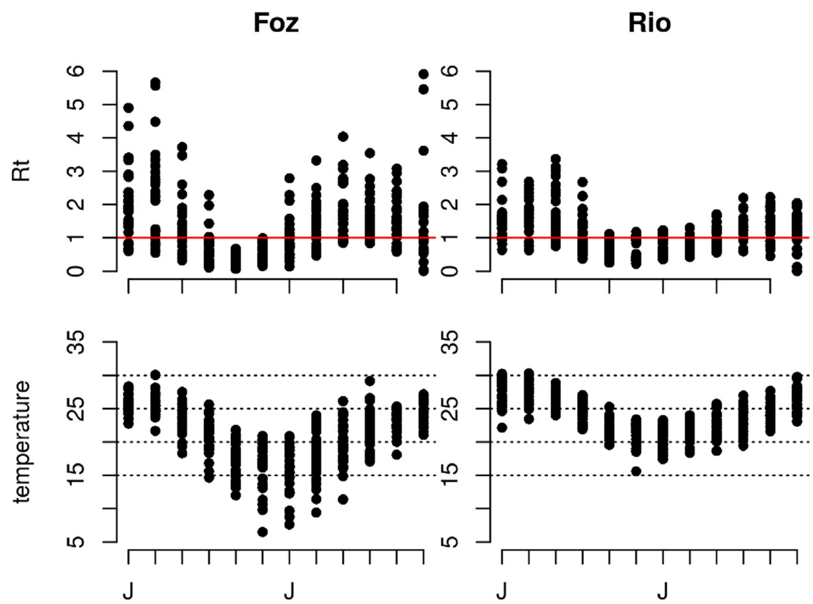

Fig. 8. (Top) Seasonality of the effective reproductive number (R_t) of dengue (J = January, July) in Foz do Iguaçu and Rio de Janeiro, Brazil (data from 2010 to 2016 shown per week); (Bottom) Distribution of mean weekly temperature, throughout the year (same period).

to our calculation, is 19 days at 30 °C, and 17 days at 33 °C. Case-to-case time intervals of up to 27 days Aldstadt et al. (2012) could be interpreted as serial transmissions as well, while the 30–34 days intervals would be unlikely according to the proposed distribution. Aldstadt et al. (2012) interpreted this longer interval as possibly a second round of transmission. In other similar study, Aldstadt (2007) found evidence for a serial interval of 18–19 days in Florida, Puerto Rico. Considering that summer temperature in this city typically varies between 25 and 31 °C, this interval is consistent with our expectations.

For the Brazilian cities investigated here, the expected generation interval of dengue according to our calculations should be 3–4 weeks in the summer months in Foz and 2.5–3 weeks in the summer months in Rio. Rio de Janeiro has a climate similar to the cities described by Aldstadt, and the generation interval is comparable. Foz, on the other hand, has a shorter and cooler transmission season. It is interesting to note that high values of R_t do not imply high epidemic peaks (Figure 5). For example, Rio de Janeiro in 2012 witnessed a large epidemic with a relatively small R_t. This was possible because the transmission season was large and there was time for the epidemic to spread, even if at a lower rate. The same occurs in Foz do Iguaçu in 2016. These results suggest that a long sustained period of transmission is responsible for large epidemics, not necessarily a high value of R_t. For surveillance, attention should be focused on the early onset of the transmission season, as an alert signal for an epidemic.

Our analysis is restricted to weekly data, because this is the temporal resolution commonly used for surveillance. However, studies have shown that daily fluctuations in temperature are important for dengue transmission (Carrington et al., 2013; Lambrechts et al., 2011) and may affect the estimation of the generation interval (Siraj et al., 2017). Although not done here, our method can easily be extended to more detailed temperature profiles. Another factor not considered is transmission during the asymptomatic period. Duong et al. (2015) showed that transmission can occur 1–2 days before the onset of symptoms. We argue that for the current purposes, which is to estimate the reproduction number of dengue using weekly data, i.e., number of cases aggregated per week, this potential 1–2 days difference does not have a significant impact. More empirical studies like Duong's are necessary to derive a probability distribution that includes the pre-symptomatic transmission phase.

A previous study by Siraj et al. (2017) derived a generation interval distribution for dengue from first principles. Our approach is similar but differ in some key aspects that are worth mentioning. First, they used lognormal distributions to represent the steps of the transmission cycle while we used gamma and exponential curves. According to Chan and Johansson (2012), the lognormal distribution presents slightly better goodness-of-fit to the available empirical data, while the gamma models comes in second. However, we opted to use the gamma and exponential distributions instead of lognormal distributions due to their mathematical properties, particularly, because there is a close link between these distributions and SIR models. For example, derive an expression for R0 using gamma and exponential functions can be done easily by combining a sequence of compartments (Lloyd, 2001) thus facilitating the derivation of other expressions for the R_t as described in Wallinga and Lipsitch (2007). In the end, numerically, the generation time distributions of the two studies are similar.

To assess the impact of temperature in the basic reproductive number of dengue, Siraj et al. (2017) use the Ross–MacDonald formulation modified to have some of its parameters as functions of temperature, that is, the mosquito density, the biting rate, the mosquito mortality and the extrinsic incubation period. They found that the basic reproduction number, $R0$, and the epidemic growth rate should vary with temperature. $R0$ in the Ross–MacDonald model rests upon assumptions of a completely susceptible population. In the present work, we used an empirical approach (Wallinga and Lipsitch, 2007) for estimating the time varying reproduction number using the weekly number of cases. This is important for surveillance when we cannot assume completely susceptible populations.

Using a simple branching process model, we illustrated how the shrink-and-expand effect of temperature on the generation interval of dengue can generate oscillations in the incidence curve. If not considered, this effect will cause inaccurate estimation of R_t. Other sources of variation of the generation interval distribution, beyond the temperature, can occur as well. During an epidemic, for example, when the number of encounters between infected and susceptibles decrease with time, the generation interval will tend to contract as the prevalence of infection increases (Svensson, 2007). Variation in mosquito abundance will also affect the generation interval, the higher the abundance of mosquitoes, the earlier the transmission is likely to occur, the higher the biting rate too. These are traits that change with temperature (Goindin et al., 2015). Although not considered in the model, the extension is possible and one possibility is taking T_H as a function of mosquito abundance (Huber et al., 2016). For cities such as Rio de Janeiro, where mosquito abundance varies not only seasonal but from area to area, the incorporation of mosquito abundance in the model should be informed by entomological data (Honorio et al., 2009).

In conclusion, this study contributes to the use of the effective reproductive number as an additional indicator in the toolkit of the public health professional, not only during the emergence of new diseases, but also during the monitoring of endemic diseases. The reproductive number is a measure of the velocity of disease spread, and as such, it marks the beginning of the epidemic, being more timely than the monitoring of incidence itself. Often in Brazil and elsewhere, disease surveillance in the presence of seasonality is carried out by comparing current incidence to incidence in the same week in previous years, using a control diagram. Although useful, this approach does not take into consideration the temporal variation of the time series. When using the reproductive number, the goal is not to compare with the last year, but with the last generation of cases, what is key to detect sustained transmission and provide alerts for contingency plans, as currently used in http://www.info.dengue.mat.br.

Author contributions statement

All authors contributed to the conception of the study and writing the manuscript. DV and CTC developed the statistical model and FCC developed the simulation model.

Competing financial interests

The authors declare no competing interests.

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References

Aldstadt, J., Yoon, I.K., Tannitisupawong, D., Jarman, R.G., Thomas, S.J., Gibbons, R.V., Uppapong, A., Iamsirithaworn, S., Rothman, A.L., Scott, T.W., Endy, T., 2012. Space-time analysis of hospitalised dengue patients in rural Thailand reveals important temporal intervals in the pattern of dengue virus transmission. Trop. Med. Int. Health 17 (9), 1076–1085.

Aldstadt, J., 2007. An incremental Knox test for the determination of the serial interval between successive cases of an infectious disease. Stoch. Environ. Res. Risk Assess 21, 487–500.

Bacaër, N., 2007. Approximation of the basic reproduction number r 0 for vector-borne diseases with a periodic vector population. Bull. Math. Biol. 69 (3), 1067–1091.

Boelle, P.Y., Thomas, G., Vergu, E., Renault, P., Valleron, A.J., Flahault, A., 2008. Investigating transmission in a two-wave epidemic of Chikungunya fever, Réunion Island. Vector Borne Zoonotic Dis. 8 (2), 207–217.

Buddhari, D., Aldstadt, J., Endy, T.P., Srikiatkhachorn, A., Thaisomboonsuk, B., Klungthong, C., Nisalak, A., Khuntirat, B., Jarman, R.G., Fernandez, S., Thomas, S.J., Scott, T.W., Rothman, A.L., Yoon, I.K., 2014. Dengue virus neutralizing antibody levels associated with protection from infection in Thai cluster studies. PLoS Negl. Trop. Dis. 8 (10), e3230.

Carrington, L.B., Armijos, M.V., Lambrechts, L., Scott, T.W., 2013. PLOS Negl. Trop. Dis. 7 (4), 1–8. http://dx.doi.org/10.1371/journal.pntd.0002190.

Cauchemez, S., Boelle, P.Y., Donnelly, C.A., Ferguson, N.M., Thomas, G., Leung, G.M., Hedley, A.J., Anderson, R.M., Valleron, A.J., 2006. Real-time estimates in early detection of SARS. Emerg. Infect. Dis. 12 (1), 110–113.

Chan, M., Johansson, M.A., 2012. The incubation periods of dengue viruses. PLoS ONE 7 (11), e50972. http://dx.doi.org/10.1371/journal.pone.0050972.

Churcher, T.S., Cohen, J.M., Novotny, J., Ntshalintshali, N., Kunene, S., Cauchemez, S., 2014. Public health. Measuring the path toward malaria elimination. Science 344 (6189), 1230–1232.

Codeco, C., Cruz, O., Riback, T.I., Degener, C.M., Gomes, M.F., Villela, D., Bastos, L., Camargo, S., Saraceni, V., Lemos, M.C.F., Coelho, F.C., 2016. InfoDengue: a nowcasting system for the surveillance of dengue fever transmission, Tech. Rep. http://dx.doi.org/10.1101/046193. biorxiv;046193v1.

Cori, A., Ferguson, N.M., Fraser, C., Cauchemez, S., 2013. A new framework and software to estimate time-varying reproduction numbers during epidemics. Am. J. Epidemiol. 178 (9), 1505–1512. http://dx.doi.org/10.1093/aje/kwt133.

Cowling, B.J., Lau, M.S., Ho, L.M., Chuang, S.K., Tsang, T., Liu, S.H., Leung, P.Y., Lo, S.V., Lau, E.H., 2010. The effective reproduction number of pandemic influenza: prospective estimation. Epidemiology 21 (6), 842–846.

Diekmann, Heesterbeek, J.A.P., 2000. Mathematical Epidemiology of Infectious Diseases: Model Building, Analysis and Interpretation. Wiley Series in Mathematical and Computational Biology. (Wiley series in mathematical and computational biology).

Duong, V., Lambrechts, L., Paul, R.E., Ly, S., Lay, R.S., Long, K.C., Huy, R., Tarantola, A., Scott, T.W., Sakuntabhai, A., Buchy, P., 2015. Asymptomatic humans transmit dengue virus to mosquitoes. Proc. Natl. Acad. Sci. U.S.A. 112 (47), 14688–14693.

Farrington, C.P., Andrews, N.J., Beale, A.D., Catchpole, M.A., 1996. A statistical algorithm for the early detection of outbreaks of infectious disease. J. R. Stat. Soc. Ser. A (Stat. Soc.) 159 (3), 547–563. http://www.jstor.org/stable/2983331.

Fine, P.E., 2003. The interval between successive cases of an infectious disease. Am. J. Epidemiol. 158 (11), 1039–1047.

Goindin, D., Delannay, C., Ramdini, C., Gustave, J., Fouque, F., 2015. Parity and longevity of *Aedes aegypti* according to temperatures in controlled conditions and consequences on dengue transmission risks. PLOS ONE 10 (8), e0135489.

Greenman, J.V., Pasour, V.B., 2011. Phase control of resonant systems: interference, chaos and high periodicity. J. Theor. Biol. 278 (1), 74–86.

Honorio, N.A., Codeco, C.T., Alves, F.C., Magalhaes, M.A., Lourenco-De-Oliveira, R., 2009. Temporal distribution of *Aedes aegypti* in different districts of Rio de Janeiro, Brazil, measured by two types of traps. J. Med. Entomol. 46 (5), 1001–1014.

Huber, J.H., Johnston, G.L., Greenhouse, B., Smith, D.L., Perkins, T.A., 2016. Quantitative, model-based estimates of variability in the generation and serial intervals of Plasmodium falciparum malaria. Malar. J. 15 (1), 490.

Kenah, E., Lipsitch, M., Robins, J.M., 2008. Generation interval contraction and epidemic data analysis. Math. Biosci. 213 (1), 71–79.

Kramer, L.D., Ebel, G.D., 2003. Dynamics of flavivirus infection in mosquitoes. Adv. Virus Res. 60, 187–232.

Lambrechts, L., Scott, T.W., Gubler, D.J., 2010. Consequences of the expanding global distribution of *Aedes albopictus* for dengue virus transmission. PLoS Negl. Trop. Dis. 4 (5), e646.

Lambrechts, L., Paaijmans, K.P., Fansiri, T., Carrington, L.B., Kramer, L.D., Thomas, M.B., Scott, T.W., 2011. Impact of daily temperature fluctuations on dengue virus transmission by *Aedes aegypti*. Proc. Natl. Acad. Sci. U.S.A. 108 (18), 7460–7465.

Lloyd, A.L., 2001. Realistic distributions of infectious periods in epidemic models: changing patterns of persistence and dynamics. Theor. Popul. Biol. 60 (1), 59–71.

Mordecai, E.A., Cohen, J.M., Evans, M.V., Gudapati, P., Johnson, L.R., Lippi, C.A., Miazgowicz, K., Murdock, C.C., Rohr, J.R., Ryan, S.J., Savage, V., Shocket, M.S., Stewart Ibarra, A., Thomas, M.B., Weikel, D.P., 2017. Detecting the impact of temperature on transmission of Zika, dengue, and chikungunya using mechanistic models. PLoS Negl. Trop. Dis. 11 (4), e0005568.

Moschopoulos, P., 1985. The distribution of the sum of independent gamma random variables. Ann. Inst. Stat. Math. 37 (1), 541–544.

Nguyen, M.N., Duong, T.H., Trung, V.T., Nguyen, T.H., Tran, C.N., Long, V.T., Dui, L.E.T., Nguyen, H.L., Farrar, J.J., Holmes, E.C., Rabaa, M.A., Bryant, J.E., Nguyen, T.T., Nguyen, H.T., Nguyen, L.T., Pham, M.P., Nguyen, H.T., Luong, T.T., Wills, B., Nguyen, C.V., Wolbers, M., Simmons, C.P., 2013. Host and viral features of human dengue cases shape the population of infected and infectious *Aedes aegypti* mosquitoes. Proc. Natl. Acad. Sci. U.S.A. 110 (22), 9072–9077.

Nishiura, H., 2010. Time variations in the generation time of an infectious disease: implications for sampling to appropriately quantify transmission potential. Math. Biosci. Eng. 7 (4), 851–869. http://dx.doi.org/10.3934/mbe.2010.7.851. http://aimsciences.org/journals/displayArticlesnew.jsp?paperID=5545.

Obadia, T., Haneef, R., Boëlle, P.-Y., 2012. The R0 package: a toolbox to estimate reproduction numbers for epidemic outbreaks. BMC Med. Inform. Decis. Mak. 12 (1). http://dx.doi.org/10.1186/1472-6947-12-147.

Sá Carvalho, M., Lima Andreozzi, V., Torres Codeço, C., Pereira Campos, D., Tereza Serrano Barbosa, M., Emiko Shimakura, S., 2011. Análise de sobrevivência: teoria e aplicaç oes em saúde. Editora Fiocruz.

Scott, T.W., Amerasinghe, P.H., Morrison, A.C., Lorenz, L.H., Clark, G.G., Strickman, D., Kittayapong, P., Edman, J.D., 2000. Longitudinal studies of *Aedes aegypti* (Diptera: Culicidae) in Thailand and Puerto Rico: blood feeding frequency. J. Med. Entomol. 37 (1), 89–101.

Simoes, T.C., Codeco, C.T., Nobre, A.A., Eiras, A.E., 2013. Modeling the non-stationary climate dependent temporal dynamics of *Aedes aegypti*. PLOS ONE 8 (8), e64773.

Siraj, A.S., Oidtman, R.J., Huber, J.H., Kraemer, M.U.G., Brady, O.J., Johansson, M.A., Perkins, T.A., 2017. Temperature modulates dengue virus epidemic growth rates through its effects on reproduction numbers and generation intervals. PLoS Negl. Trop. Dis. 11 (7), e0005797.

Smith, C.E., 1975. The significance of mosquito longevity and blood-feeding behaviour in the dynamics of arbovirus infections. Med. Biol. 53 (5), 288–294.

Svensson, A., 2007. A note on generation times in epidemic models. Math. Biosci. 208 (1), 300–311.

Vezzani, D., Velazquez, S.M., Schweigmann, N., 2004. Seasonal pattern of abundance of *Aedes aegypti* (Diptera: Culicidae) in Buenos Aires City, Argentina. Mem. Inst. Oswaldo Cruz 99 (4), 351–356.

Villela, D.A., Codeco, C.T., Figueiredo, F., Garcia, G.A., Maciel-de Freitas, R., Struchiner, C.J., 2015. A Bayesian hierarchical model for estimation of abundance and spatial

density of *Aedes aegypti*. PLOS ONE 10 (4), e0123794.
Vink, M.A., Bootsma, M.C., Wallinga, J., 2014. Serial intervals of respiratory infectious diseases: a systematic review and analysis. Am. J. Epidemiol. 180 (9), 865–875.
Wallinga, J., Lipsitch, M., 2007. How generation intervals shape the relationship between growth rates and reproductive numbers. Proc. Biol. Sci. 274 (1609), 599–604.
Wallinga, J., Teunis, P., 2004. Different epidemic curves for severe acute respiratory syndrome reveal similar impacts of control measures. Am. J. Epidemiol. 160 (6), 509–516.
Yang, H.M., Macoris, M.L., Galvani, K.C., Andrighetti, M.T., Wanderley, D.M., 2009. Assessing the effects of temperature on dengue transmission. Epidemiol. Infect. 137 (8), 1179–1187.